

What are Abstract Methods?

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- ▶ It does not contain any body.
- ▶ Abstract method has simply a **signature declaration followed by a semicolon**.
- ▶ Since the **abstract method does not contain any body**.
 - ★ Therefore, it is also known as **incomplete method in java**.
- ▶ **Key point:** A **concrete method** is a **method** which has **always the body**.
 - ★ It is also called a **complete method in java**.

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What are Abstract Methods?

- Syntax:

```
abstract type MethodName(arguments); // No body  
For example:  
    abstract void msg(); // No body.
```

- An abstract class can have **both abstract and regular methods**:

```
abstract class Animal  
{  
    public abstract void animalSound();  
    public void sleep()  
    {  
        System.out.println("Zzz");  
    }  
}
```

When to use Abstract method in Java?

- ▶ There are the following uses of abstract method in Java. They are as follows:
 - 1 An abstract method can be used when the same method has to perform different tasks depending on the object calling it.
 - 2 A method can be used as abstract when you need to be overridden in its non-abstract subclasses.

Abstract Class

Example: Java Abstract Class

```
abstract class Language {  
    public void display() {  
        System.out.println("This is Java Programming");  
    }  
}  
  
class Main extends Language {  
  
    public static void main(String[] args) {  
        Main obj = new Main();  
        obj.display();  
    }  
}
```



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Abstract Class

Example: Java Abstract Class

```
abstract class Shape{
    abstract void draw();
}
class Rectangle extends Shape{
    void draw(){System.out.println("drawing rectangle");}
}
class Circle1 extends Shape{
    void draw(){System.out.println("drawing circle");}
}
class TestAbstraction1{
    public static void main(String args[]){
        Shape s=new Circle1();
        s.draw();
    }
}
```



Abstract Class

Example: Java Abstract Class

```
abstract class AbstractDemo{
    public void myMethod(){
        System.out.println("Hello");
    }
    abstract public void anotherMethod();
}

public class Demo extends AbstractDemo{

    public void anotherMethod() {
        System.out.print("Abstract method");
    }
    public static void main(String args[])
    {

        AbstractDemo obj = new AbstractDemo();
        obj.anotherMethod();
    }
}
```



Abstract Class

Example: Java Abstract Class

```
abstract class Bank{
    abstract int getRateOfInterest();
}
class SBI extends Bank{
    int getRateOfInterest(){return 7;}
}
class PNB extends Bank{
    int getRateOfInterest(){return 8;}
}
class Main{
    public static void main(String args[]){
        Bank b;
        b=new PNB();
        System.out.println("Rate of Interest is: "+b.getRateOfInterest()+" %");
        b=new SBI();
        System.out.println("Rate of Interest is: "+b.getRateOfInterest()+" %");
    }}

```



Reading Assignment

- ❶ What is the difference between Abstract and Encapsulation?
- ❷ What is Interface?
- ❸ What is the difference between Abstract and Interface?

► End of Chapter Five